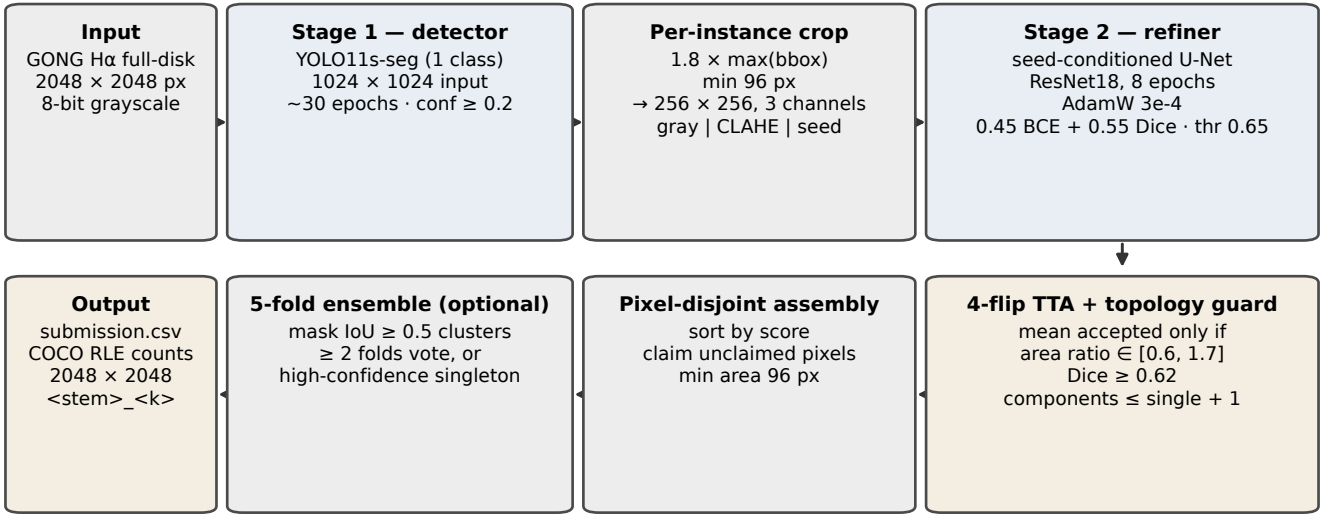
Input
GONG Hα full-disk
2048 × 2048 px
8-bit grayscale
Stage 1 — detector
YOLO11s-seg (1 class)
1024 × 1024 input
~30 epochs · conf ≥ 0.2
Per-instance crop
1.8 × max(bbox)
min 96 px
→ 256 × 256, 3 channels
gray | CLAHE | seed
Stage 2 — refiner
seed-conditioned U-Net
ResNet18, 8 epochs
AdamW 3e-4
0.45 BCE + 0.55 Dice · thr 0.65
4-flip TTA + topology guard
mean accepted only if
area ratio ∈ [0.6, 1.7]
Dice ≥ 0.62
components ≤ single + 1
Pixel-disjoint assembly
sort by score
claim unclaimed pixels
min area 96 px
5-fold ensemble (optional)
mask IoU ≥ 0.5 clusters
≥ 2 folds vote, or
high-confidence singleton
Output
submission.csv
COCO RLE counts
2048 × 2048
<stem>_<k>